

Loss Landscape Topology Reveals Why Simple Baselines are Competitive at 3D Point Cloud Segmentation Under Class Imbalance



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TL;DR: In point-based 3D segmentation, plain cross-entropy is a robust baseline. The loss landscape, influenced by imbalance severity, offers limited opportunities for specialized losses to improve performance.

11

imbalance-mitigation
methods evaluated

0.8-3.3%

mIoU gap: CE vs. the best
specialized method

641:1 & 56:1

imbalance ratios
(DALES / S3DIS)

2

point-based architectures
(KPConv, RandLA-Net)

Motivation

- LiDAR point-cloud segmentation drives autonomous navigation, robotics, and environmental monitoring.
- Real LiDAR is severely long-tailed: a few majority classes dominate, while safety-critical rare classes (signs, people) hold <0.1% of points.
- Missing these rare classes is precisely where the safety cost is highest.
- The natural fix is to import 2D imbalance losses (proven on CIFAR-LT, ImageNet-LT, LVIS).

Question & Contributions

Do specialized imbalance losses that help in 2D actually help point-based 3D segmentation (and if not, why?)

We contribute:

- Systematic evaluation of 11 methods on two datasets of contrasting imbalance.
- Confusion-matrix analysis exposing precision-recall trade-off that keeps aggregate mIoU flat.
- Decision-boundary variability revealing dataset-dependent landscape topology.
- Loss-landscape characterization via weight perturbation + Hessian spectra.

Experimental Setup

Reweighting schemes (6)

uniform (CE) · inverse-frequency · class-balanced · inverse-log · inverse-power · complementary-frequency

Loss functions (5)

Focal · LDAM · Logit-Adjustment · Balanced-Softmax · Seesaw

DALES

aerial (8 classes)
641:1 imbalance

S3DIS

indoor (13 classes)
56:1 imbalance

Three analyses

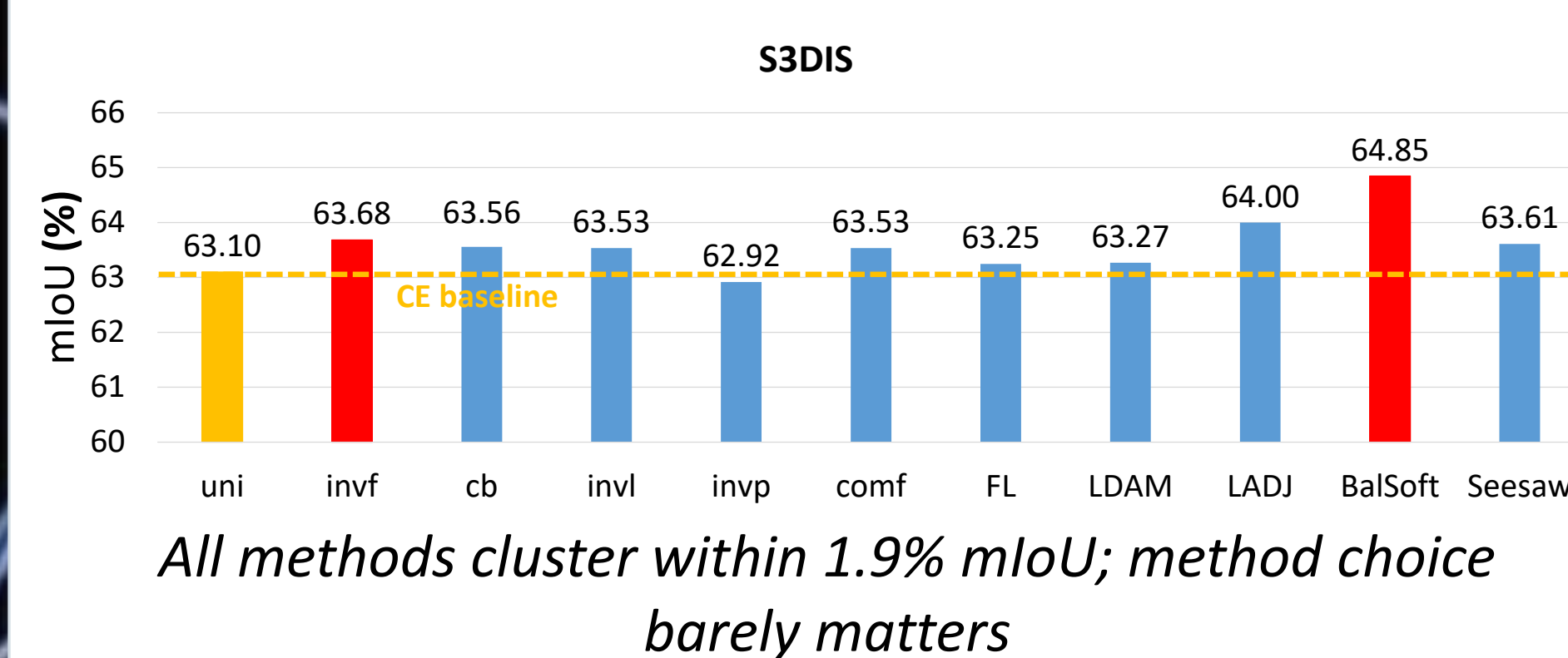
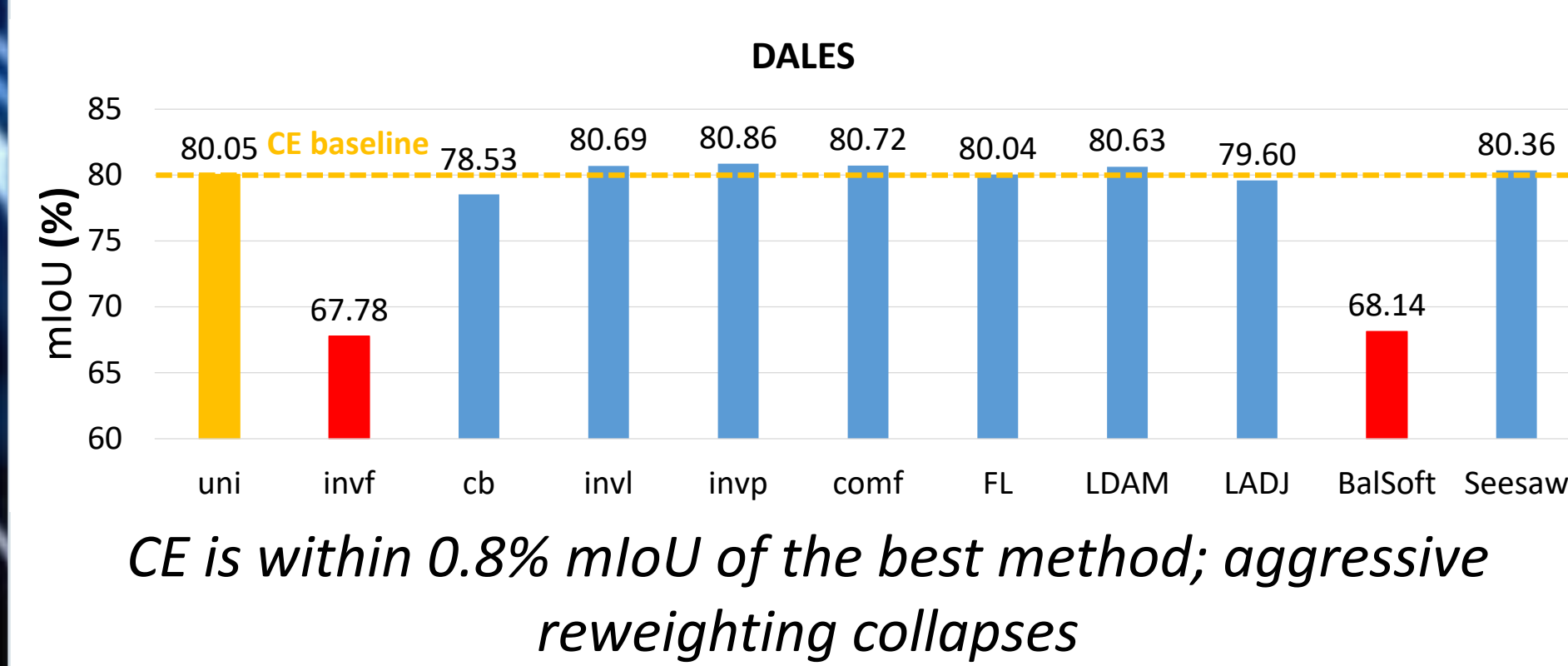
- Confusion matrix → per-class precision / recall
- Decision-boundary variability vs. CE
- Loss-landscape topology: weight perturbation + top-10 Hessian eigenvalues

KPConv & RandLA-Net

per-class IoU / mIoU with 10 voting passes

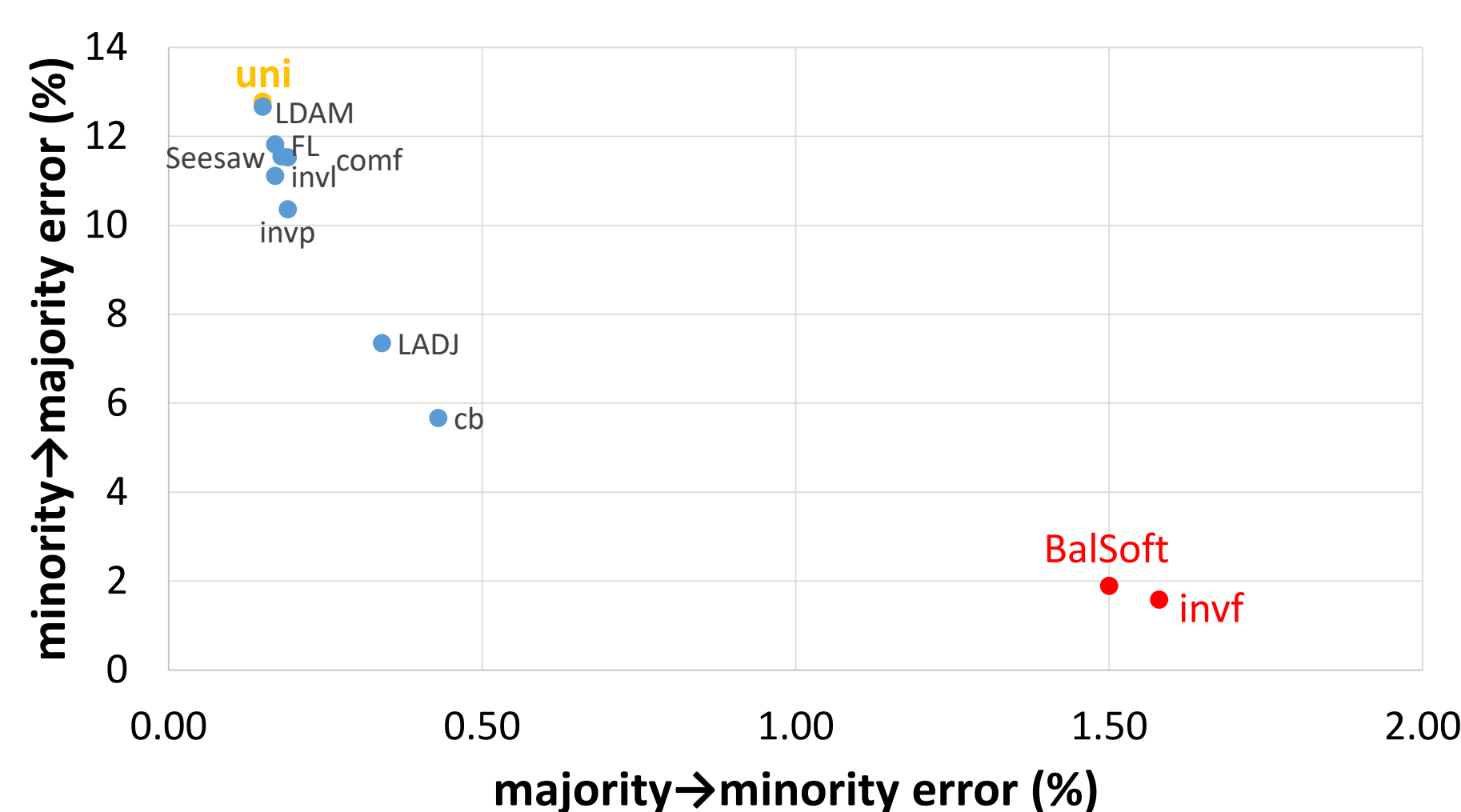
CE is Competitive

- Standard CE is within 0.8-3.3% mIoU of every specialized method. Under extreme imbalance the aggressive schemes collapse.



Why mIoU Stays Flat

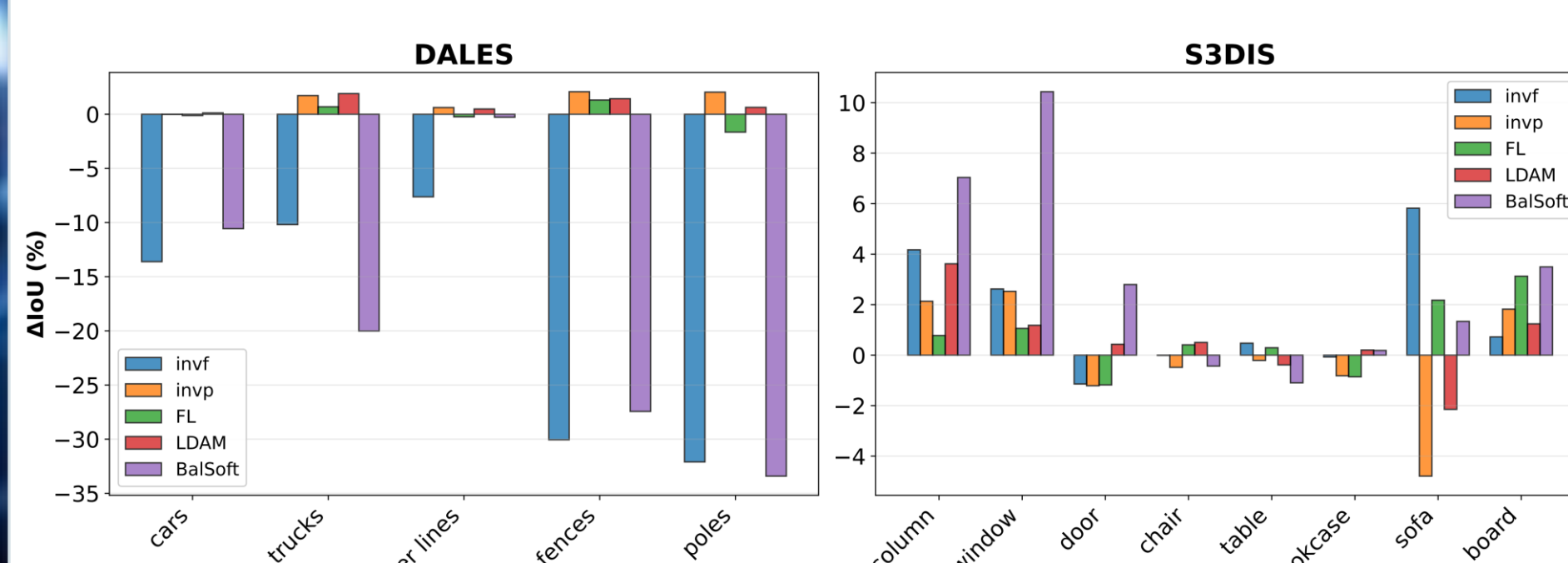
- Specialized losses raise minority recall (aggressive reweighting cuts minority → majority errors up to 10x). But they inflate majority → minority errors by the same factor, so precision collapses and aggregate mIoU barely moves.



Net effect: recall gains and precision losses cancel

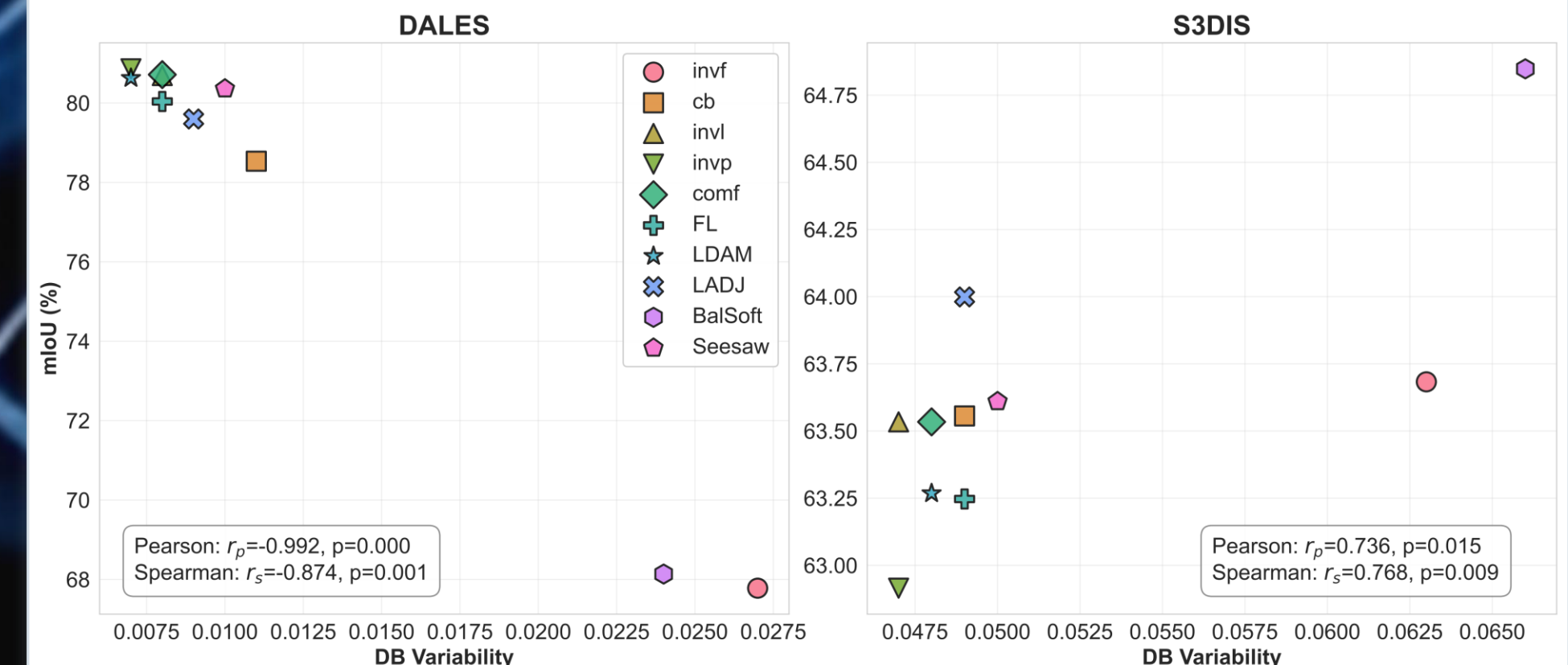
Minority Classes Don't Benefit

- Per-class ΔIoU vs. CE on rare classes (<6% frequency): aggressive reweighting degrades DALES minorities; on S3DIS all methods show much smaller differences, so method choice has little impact.



Loss-Landscape Topology

- Decision-boundary variability vs. CE tells two different stories (set by imbalance severity, not by



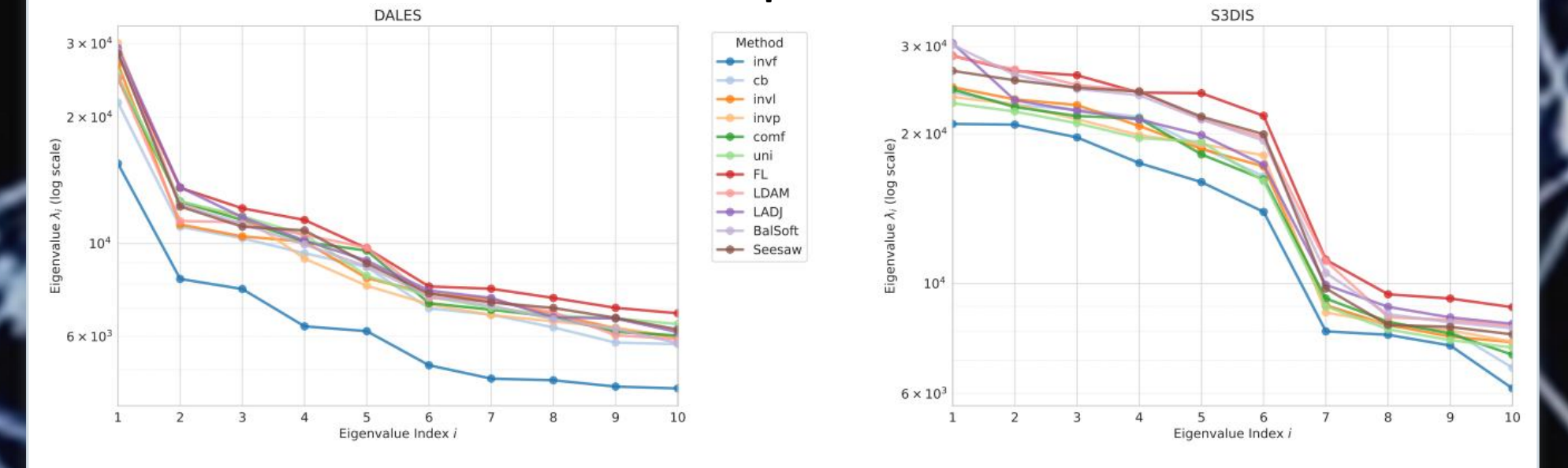
DALES

deviating boundaries
cost >12%

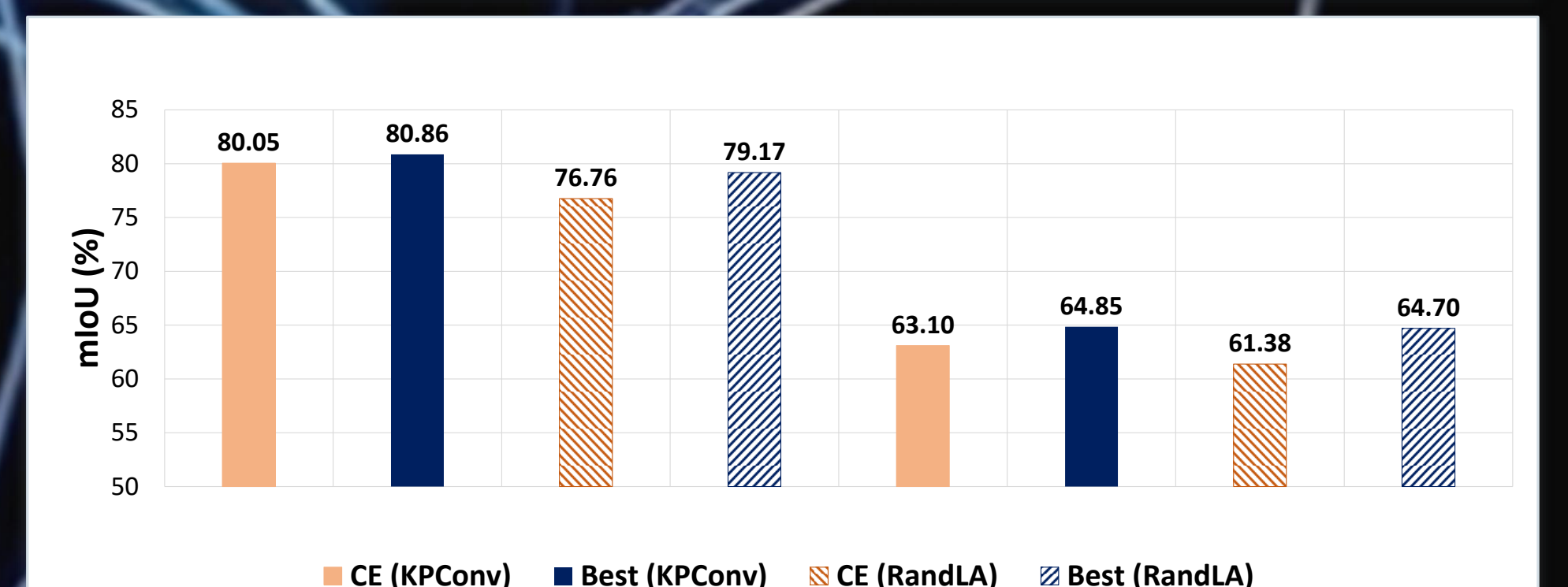
S3DIS

methods scatter 0.047-0.066
variability; stay within ≈2%

- top-10 Hessian eigenvalues cluster within each dataset; sharp / anisotropic under extreme imbalance; flat / isotropic under moderate.



Across Architectures



- Same story on KPConv and RandLA-Net despite opposite sampling strategies
- Finding is not architecture-specific (KPConv gains 0.8-1.75%; RandLA-Net 2.4-3.3%).

Takeaways & Future Work

- Adopt CE with confidence; reserve specialized losses for careful, per-dataset tuning (not as a default).
- Specialized losses give modest, architecture-dependent gains, and risk severe degradation if mis-tuned.
- Imbalance severity, not loss choice, shapes the optimization landscape; neighbourhood aggregation may already provide implicit class balancing.

FUTURE WORK

- Extend to voxel- and transformer-based backbones (Cylinder3D, Point Transformers).
- Investigate combined data-level + loss-level strategies and landscape-aware, per-dataset method selection.